AI Shopping Assistant

An intelligent AI-powered shopping assistant using LangGraph, LangChain, FastAPI, Streamlit, and FAISS

Project Overview

The AI Shopping Assistant is an intelligent conversational shopping application that enables users to search for products using natural language instead of manually browsing large product catalogs.

The application combines Large Language Models (LLMs) with semantic search and recommendation engines to provide personalized shopping assistance. It supports multi-turn conversations, allowing users to refine their shopping preferences such as budget, brand, category, and purpose.

The project follows a modular architecture where the frontend, backend, AI workflow, and recommendation engines are separated for scalability and maintainability.

Objective

The primary objective is to simplify online shopping by allowing users to interact with an AI assistant that understands shopping intent and provides relevant recommendations.

- Understand natural language shopping queries
- Recommend products based on user preferences
- Compare products
- Generate shopping bundles
- Maintain conversational context
- Reduce manual searching effort

Features

- **AI-powered conversational assistant** — Natural language interface for shopping
- **Semantic product search** — FAISS-powered vector similarity search
- **Personalized recommendations** — Context-aware product suggestions
- **Product comparison** — Side-by-side attribute comparison
- **Shopping bundle generation** — Complementary product suggestions
- **Multi-turn conversation** — Persistent context across queries
- **Structured query understanding** — Intent, budget, brand extraction
- **FastAPI REST API** — Scalable backend service

- **Streamlit frontend** — Interactive web interface
- **Multi-LLM support** — Groq and Gemini compatibility

Technology Stack

Technology	Purpose
Python	Programming Language
Streamlit	Frontend UI
FastAPI	REST API Backend
LangGraph	AI Workflow
LangChain	LLM Orchestration
FAISS	Semantic Search
Sentence Transformers	Text Embeddings
Pandas	Data Processing
Pydantic	Data Validation

Workflow

1. User enters a shopping query
2. Streamlit sends the request to FastAPI
3. FastAPI forwards the query to the Shopping Agent
4. Shopping Graph processes the request
5. Query Understanding extracts: Intent, Category, Budget, Brand, Purpose, Constraints
6. The graph routes the query to Recommendation, Comparison, or Bundle Engine
7. FAISS performs semantic search on the product dataset
8. The selected engine prepares the response
9. FastAPI returns the result
10. Streamlit displays recommendations in the interface

Module Documentation

Shopping Agent

Acts as the central controller of the application. It receives user queries, invokes the Shopping Graph, and returns the generated response.

Shopping Graph

Coordinates the AI workflow by routing requests to the appropriate engine based on the detected intent.

Query Understanding Engine

Uses structured LLM output to identify intent, product category, budget, brand, purpose, constraints, and product names.

Recommendation Engine

Retrieves relevant products using semantic search and ranks them based on the user's shopping requirements.

Comparison Engine

Compares multiple products based on available attributes and presents a structured comparison.

Bundle Engine

Suggests complementary products to create shopping bundles.

Semantic Search

Uses FAISS and sentence-transformer embeddings to retrieve semantically similar products.

API Documentation

POST /chat

Request:

```
{ "query": "Recommend a laptop under Rs.60000" }
```

Response:

```
{ "response": "...", "provider": "Groq", "shopping_context": {}, "products": [],
"comparison": {}, "bundle": {} }
```

Run Instructions

Backend

```
uvicorn api.app:app --reload
```

Swagger UI: http://127.0.0.1:8000/docs

Frontend

```
streamlit run frontend/app.py
```

Example Queries

- Recommend a laptop under Rs.60000

- Compare iPhone 16 and Samsung Galaxy S25
- Best earbuds under Rs.5000
- Create a gym shopping bundle
- Suggest a black shirt for Goa trip

Future Scope

- User authentication
- Shopping cart integration
- Voice-based interaction
- Real-time pricing APIs
- Multi-language support
- Personalized user profiles
- Order tracking
- Cloud deployment

Conclusion

The AI Shopping Assistant demonstrates how conversational AI, semantic search, and recommendation systems can be integrated into a modern shopping platform. By combining LangGraph, FastAPI, Streamlit, and FAISS, the project provides a scalable and modular architecture that enables intelligent product discovery and enhances the overall shopping experience.